

PROJECT PLANNING PHASE

ComicCraft – AI Comic Story Creator using Gemini Models

1. Planning Overview

The Project Planning Phase organizes the development of ComicCraft into clear milestones and activities. The plan covers model selection and architecture, core functionality development, FastAPI route development, frontend development, and deployment.

2. Main Planning Goals

The project plan is structured to achieve the following outcomes:

- Select suitable generative AI models for storytelling and image creation.
- Define the overall application architecture and development environment.
- Implement the core comic-generation functions.
- Build FastAPI routes for user input and AI workflow management.
- Develop the web interface using HTML, CSS, and Jinja2.
- Test the complete workflow through local deployment.

3. Development Strategy

The project follows a milestone-based development strategy. Each milestone groups related activities so that the application can progress from planning and model selection to implementation, interface development, and local deployment.

4. Project Workflow

The planned workflow moves from user input to AI-generated content and finally to comic export: user input → structured outline → story and dialogue → panel images → layout → PDF export.

PROJECT PLANNING PHASE – MILESTONE 1

Milestone 1: Model Selection and Architecture

The first milestone establishes the technical foundation of ComicCraft. It includes research and selection of appropriate generative AI models, application architecture definition, and development environment setup.

Activity 1.1 – Research and Select the Appropriate Generative AI Model

The planning activity identifies the content that ComicCraft must generate: structured comic outlines, full narration and character dialogues, and visual comic-style illustrations.

Planned evaluation factors:

- Creative writing quality for narration and dialogue.
- Prompt adherence and richness of scene descriptions.
- Image generation speed and resolution.
- Model stability, documentation, and community support.

Planned model roles:

Model	Planned Use
Gemini Flash	Fast, structured 5-panel comic outline generation.
Gemini Pro	Detailed narration and character dialogue generation.
Stable Diffusion	Comic-style illustration generation.

Activity 1.2 – Define the Application Architecture

The planning design separates ComicCraft into Frontend, Backend, and AI Integration components. The frontend captures user preferences, FastAPI processes requests, and AI services generate the story content and illustrations.

Activity 1.3 – Set Up the Development Environment

The plan includes Python and pip installation, creation of a virtual environment, installation of required libraries, and setup of the project structure. The application is planned to run through Uvicorn using the FastAPI application entry point.

PROJECT PLANNING PHASE – MILESTONE 2

Milestone 2: Core Functionalities Development

The second milestone plans the implementation of the core AI and comic-processing functions. These functions form the central generation pipeline.

Activity 2.1 – Develop the Core Functionalities

Planned Function	Module	Planned Output
generate_outline()	gemini_flash.py	Structured 5-panel outline.
generate_story()	gemini_pro.py	Narration and character dialogues.
generate_image()	image_generator.py	Comic illustration for each panel.
build_comic_layout()	layout_builder.py	Matched panel images and text.
save_pdf()	exporters.py	Multi-page downloadable PDF.

Planned Processing Sequence

1. Receive the user's story prompt and preferences.
2. Generate a structured five-panel outline.
3. Expand the outline into narration and dialogues.
4. Generate an image for every panel.
5. Organize images and story text into a layout.
6. Compile the final comic into a PDF.

Activity 2.2 – Implement the FastAPI Backend

The backend plan includes processing HTML form inputs and JSON requests, validating data through Pydantic for API-based generation, calling the AI functions, and passing the results to the frontend templates.

Planned API workflows include /generate, /generate-comic/json, and /test-image.

PROJECT PLANNING PHASE – MILESTONES 3 AND 4

Milestone 3: routes.py Development

The third milestone focuses on the main application logic in routes.py. The route handlers connect user interactions with the comic generation functions and frontend templates.

Planned Routes

- / — load the homepage.
- /generate — process form data and generate the comic preview.
- /generate-comic/json — accept JSON and return layout and PDF information.
- /export-success — display the export confirmation.
- /test-image — test image generation independently.

Milestone 4: Frontend Development

The fourth milestone plans a responsive and clean web interface using HTML, CSS, and Jinja2. The interface is organized into three main templates.

Template	Planning Purpose
index.html	Collect story prompt, character name, setting, tone, and art style.
comic_preview.html	Display generated panels, images, descriptions, captions, and narration.
export_success.html	Confirm successful comic export and provide return navigation.

Frontend Planning Activities

- Create semantic HTML form fields for all user inputs.
- Apply consistent CSS styling, spacing, buttons, and result panels.
- Use Jinja2 loops and variables to render generated panel content dynamically.
- Maintain a direct flow from comic creation to preview and export confirmation.

PROJECT PLANNING PHASE – MILESTONE 5 AND COMPLETION PLAN

Milestone 5: Deployment

The fifth milestone plans local deployment and verification of the complete ComicCraft workflow. The objective is to ensure that AI text generation, image generation, layout construction, and PDF export work together before considering future cloud deployment.

Activity 5.1 – Prepare for Local Deployment

Planned tasks include:

- Create and activate a Python virtual environment.
- Install dependencies from requirements.txt.
- Configure Gemini and Hugging Face credentials through environment variables.
- Configure static files and Jinja2 templates.

Activity 5.2 – Testing and Verification

The plan uses Uvicorn to start the FastAPI server and a browser to verify the homepage. FastAPI's interactive /docs interface is planned for testing API endpoints such as /generate-comic/json, /generate, /export-success, and /test-image.

Project Completion Checklist

- ✓ Model selection and architecture defined.
- ✓ Development environment prepared.
- ✓ Core AI generation functions implemented.
- ✓ FastAPI routes connected to the generation workflow.
- ✓ Frontend templates designed and dynamically rendered.
- ✓ Local deployment and workflow verification planned.

Final Planning Outcome

The Project Planning Phase provides a structured path from initial model selection through application development and local verification. By dividing the work into five milestones, ComicCraft can be developed as a sequence of connected tasks while keeping the AI, backend, frontend, and export components modular.

The project document also identifies future expansion possibilities, including user profiles, comic libraries, and multi-page story arcs. These can be considered after the core single-comic workflow has been completed and verified.

Conclusion

The planning phase establishes the development sequence, responsibilities, technical components, testing activities, and deployment preparation needed to build ComicCraft systematically.